



CyberBlueSOC

Your Blue Team Security Lab

Installation & User Guide



WAZUH



SURICATA



Arkime



Velociraptor



MISP



TheHive



CORTEX



Shuffle



CyberChef



CALDERA

MITRE
ATT&CK
NAVIGATOR



FleetDM



OSquery



OpenSearch



EveBox



Portainer



Table of Contents

Welcome to CyberBlueSOC.....	4
Your Complete Security Operations Center.....	5
The Problem It Solves	6
Who This Is For.....	6
Your 15+ Security Tools.....	7
Quick Installation	8
Simple Commands:.....	8
System Requirements	8
Installation Process.....	8
Portal Interface Overview	9
Tool Cards Explained.....	12
Viewing Tool Credentials.....	13
Viewing Tool Information.....	13
Tool Access Guide.....	14
All Your Tools (No Portal Password!)	14
Complete Tool List	14
Verify the Access to Tools.....	15
Velociraptor.....	15
Wazuh	15
Shuffle.....	16
MISP	16
CyberChef	17
TheHive.....	17
Cortex	18
Fleet.....	18
Arkime	19
Caldera	19
EveBox	20
Wireshark	20
ATT&CK Navigator.....	21



Portainer	21
Deploying Agents	22
Agent Deployment Center	22
Velociraptor (DFIR Agent)	22
Wazuh (SIEM Agent)	23
Arkime Network Capture (PCAP)	24
Threat Intelligence	25
IOC Search Interface	25
MISP Statistics Dashboard	26
Detection Rules	27
YARA Malware Detection	27
Sigma Detection Rules	29
System Monitoring	30
Metrics Dashboard	30
Troubleshooting	31
Quick Reference	31
Essential Commands	31
System Locations	31
Common Issues & Solutions	32
Installation Problems	32
Port Already in Use	32
Out of Disk Space	32
Docker Permission Denied	32
Runtime Issues	33
Container Keeps Restarting	33
Portal Not Loading	33
Tool Shows "Stopped"	34
Performance Issues	34
High Memory Usage	34
MISP Issues	35
Can't Login to MISP	35
Network Issues	35
Can't Access Tools from Remote	36



Table of Figures

Figure 1: CyberBlueSOC Dashboard	9
Figure 2: CyberBlueSOC Agents Deployment Dashboard.....	10
Figure 3: CyberBlueSOC Intel Dashboard.....	10
Figure 4: CyberBlueSOC Rules Dashboard.....	11
Figure 5: CyberBlueSOC Metrics Dashboard	11
Figure 6: Tools Cards.....	12
Figure 7: Tools Credentials.....	13
Figure 8: Tools Information feature	13
Figure 9: Velociraptor WebUI.....	15
Figure 10: Wazuh WebUI.....	15
Figure 11: Shuffle WebUI.....	16
Figure 12: MISP WebUI.....	16
Figure 13: CyberChef WebUI.....	17
Figure 14: TheHive WebUI)	17
Figure 15: Cortex WebUI	18
Figure 16: FleetDM WebUI.....	18
Figure 17: Arkime WebUI	19
Figure 18: MITRE Caldera WebUI.....	19
Figure 19: EveBox WebUI.....	20
Figure 20: Wireshark WebUI	20
Figure 21: MITRE ATT&CK Navigator WebUI	21
Figure 22: Portainer WebUI.....	21
Figure 23: Velociraptor Agent Deployment.....	22
Figure 24: Wazuh Agent Deployment.....	23
Figure 25: Arkime PCAP Generator.....	24
Figure 26: MISP IOC Search	25
Figure 27: MISP statistics.....	26
Figure 28: YARA Rules Categories browser	27
Figure 29: YARA Rules Browser.....	28
Figure 30: YARA Rule Content.....	28
Figure 31: Sigma Rules Categories	29
Figure 32: Sigma Rule Content	29
Figure 33: CyberBlueSOC Metrics Dashboard	30



Welcome to CyberBlueSOC

Your Complete Security Operations Center

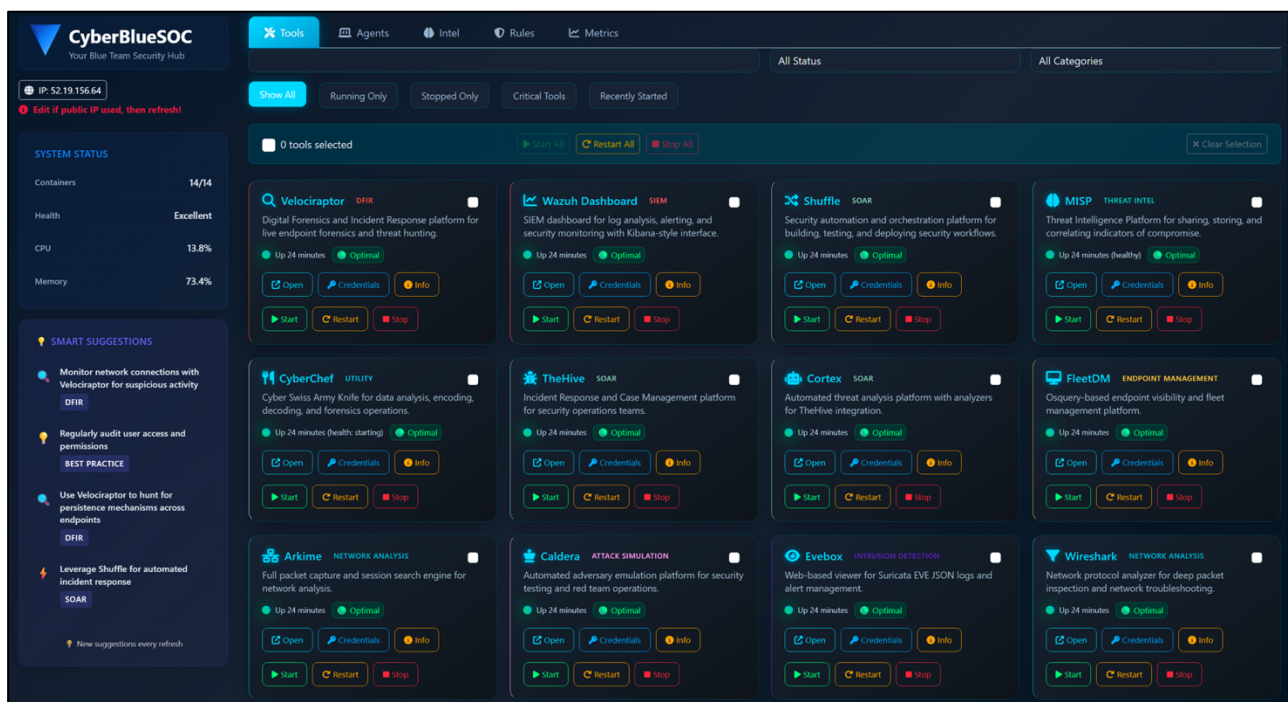
What is CyberBlueSOC?

CyberBlueSOC transforms your Ubuntu machine into a SOC Blue Team LAB in less than an hour - with the same enterprise tools used by Security Operations Centers and Blue Team Labs.

No complexity . No licenses

What You Get:

- Enterprise SIEM platform (Wazuh)
- DFIR – EDR Endpoint investigation (Velociraptor)
- Incident response platform (TheHive & Cortex)
- Security automation (Shuffle)
- Attacks Simulation (Caldera)
- 250,000+ threat indicators (MISP)
- Network forensics tools (Arkime, Wireshark)
- + additional professional tools
- Beautiful web portal - no password required





The Problem It Solves

Traditional security tool deployment:

- Takes weeks to configure
- Requires licenses
- Tools don't integrate
- Endless troubleshooting

CyberBlueSOC makes it:

- About 30-minute setup
- Using open-source
- integrated & Works out-of-the-box.

Who This Is For

- Security Students → Build job-ready skills SOC Analysts → Practice with real tools
- Career Changers → Get hands-on experience IT Professionals → Add security expertise



Your 15+ Security Tools

SIEM & Monitoring:

- **Wazuh** - Enterprise SIEM dashboard
- **Suricata** - Network intrusion detection
- **EveBox** - Real-time security events

Digital Forensics:

- **Velociraptor** - Endpoint investigation
- **Arkime** - Network traffic analysis
- **Wireshark** - Protocol analyzer

Threat Intelligence:

- **MISP** - 250,000+ threat indicators
- **MITRE ATT&CK** - Threat modeling

Incident Response:

- **TheHive** - Case management
- **Cortex** - Artifact analysis
- **Shuffle** - Security automation

Attacks Simulations

- **Caldera** - Adversary emulation

Utilities:

- **CyberChef** - Data analysis
- **Fleet** - Endpoint management
- **Portainer** - Container management

Threat Hunting:

- **YARA** - 523+ malware detection rules
- **Sigma** - 3,047+ SIEM detection rules



Quick Installation

Simple Commands:

Installation Commands

```
git clone https://github.com/CyberBlu3s/CyberBlue.git
cd CyberBlue
chmod +x cyberblue_install.sh
./cyberblue_install.sh
```

That's it! The installer runs completely automatically for 30-45 minutes.

System Requirements

Recommended Specifications

CPU: 8+cores (Intel/AMDx86_64)

RAM: 16GB minimum

Disk: 150GB+ SSD **OS:** Ubuntu22.04+

Runs on: · VMware · VirtualBox · Cloud VMs (AWS) · etc

Installation Process

The fully automated installer handles everything:

Phase 1 (10-15 min): System setup

- Installing Docker & Docker Compose
- Configuring 8GB swap space
- Installing YARA (523+ rules) & Sigma (3,047+ rules)
- System optimizations

Phase 2 (15-20 min): Deployment

- Building 30+ containers
- Configuring networks and Generating SSL certificates
- Downloading agent binaries

Phase 3 (5-10 min): Configuration

- Enabling auto-start on reboot
- Final health checks



Grab a coffee and watch the automated installation!



Your CyberBlueSOC Portal

Instant Access - No Password!

After installation, simply open your

browser: https://YOUR_SERVER_IP:5443

SSL Certificate Warning

You'll see a security warning about the certificate. This is normal for lab environments. Click **Advanced** → **Proceed** (safe in your isolated lab)

No login required! The portal opens directly - perfect for quick lab access.

Portal Interface Overview

Your portal has 5 main tabs visible at the top:

1. **Tools Tab** (Default view)
 - Manage all 15+ security tools
 - Start/Stop/Restart controls
 - View credentials for each tool

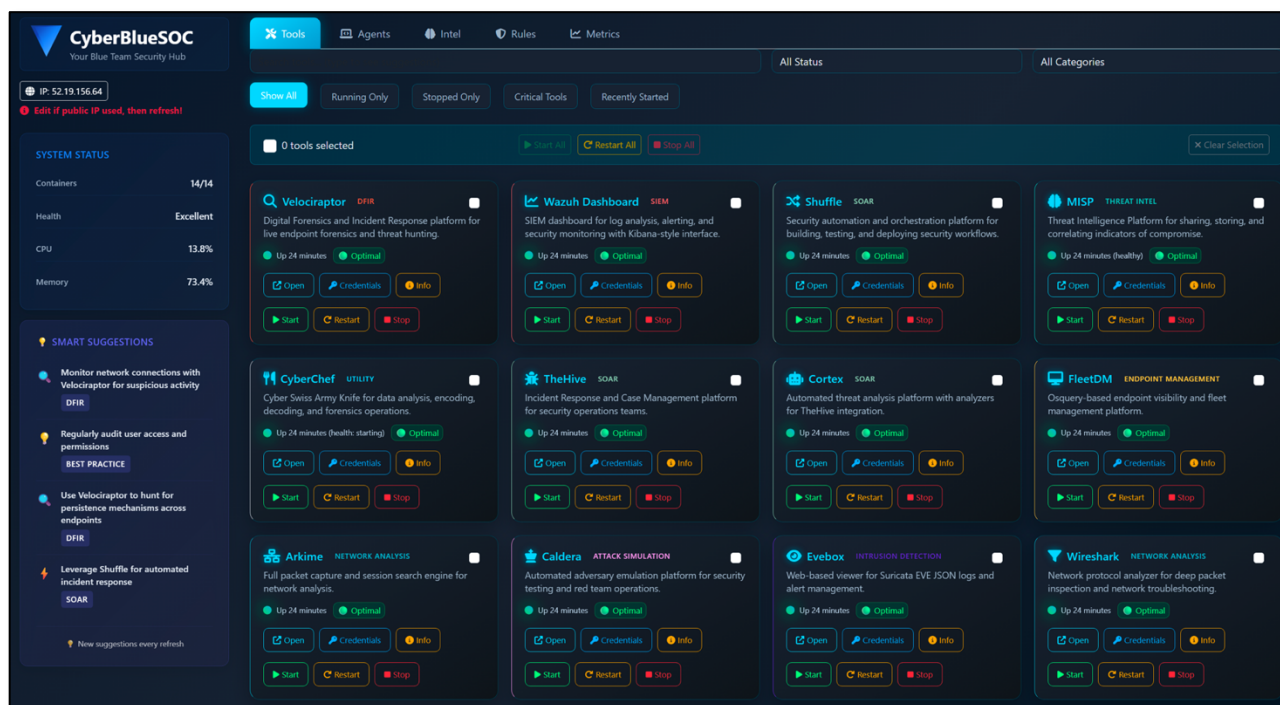


Figure 1: CyberBlueSOC Dashboard

2. Agents Tab

- Deploy Velociraptor to Windows/Linux/macOS
- Deploy Wazuh agents
- Start network captures with Arkime

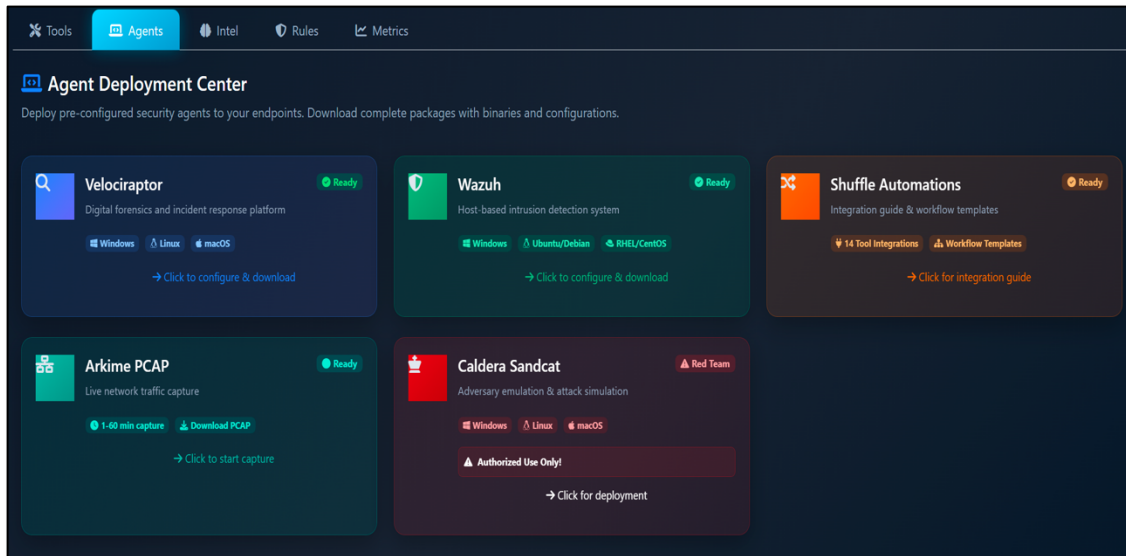


Figure 2: CyberBlueSOC Agents Deployment Dashboard

3. Intel Tab

- Search 200,000+ threat indicators
- Query IPs, domains, hashes
- View MISP statistics
- See recent threat events

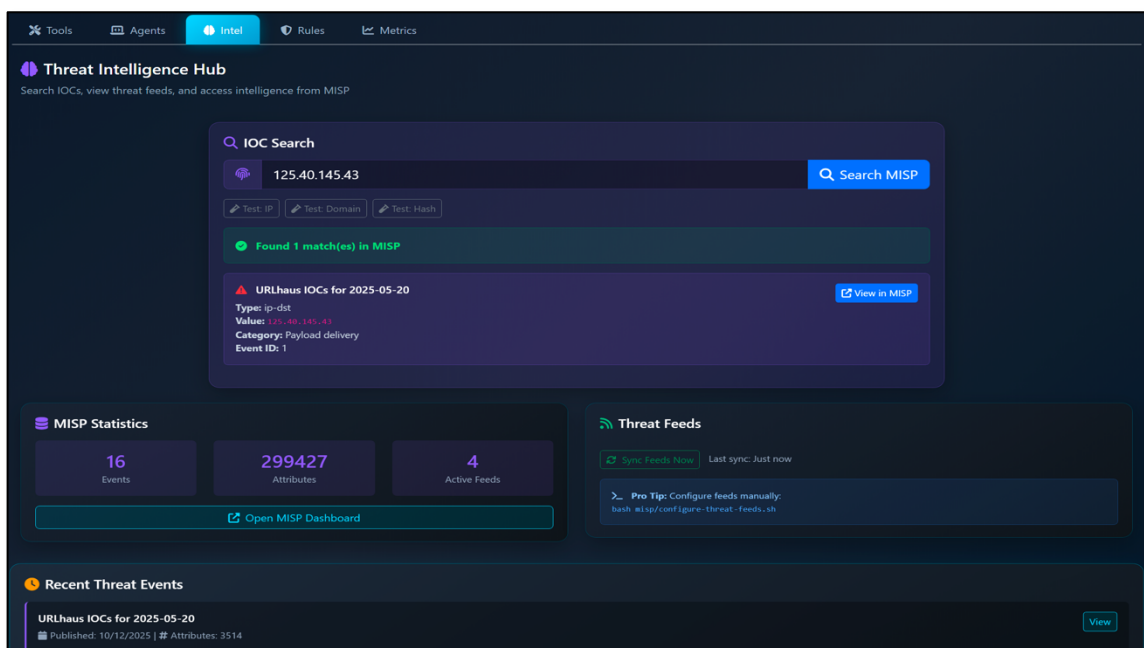


Figure 3: CyberBlueSOC Intel Dashboard



4. Rules Tab

- Browse 523 YARA malware rules
- Browse 3,047 Sigma detection rules
- View rule content
- Copy rules to clipboard

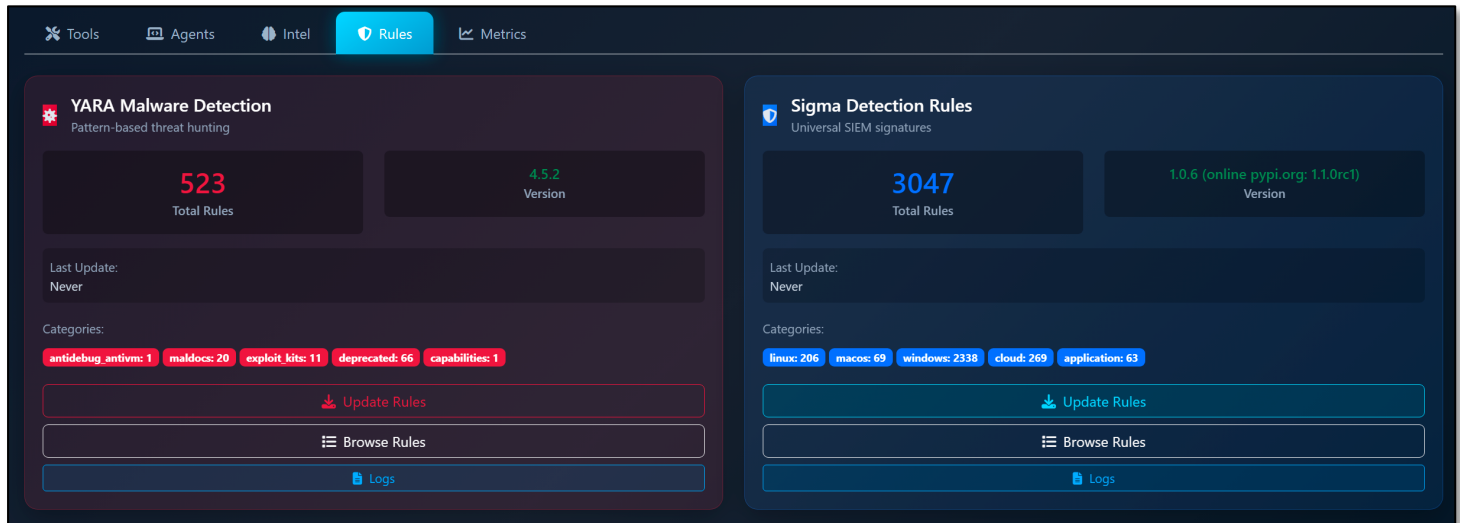


Figure 4: CyberBlueSOC Rules Dashboard

5. Metrics Tab

- System performance dashboard
- Container health monitoring
- Network statistics
- Recent events
- Visual grid showing all 30+ containers

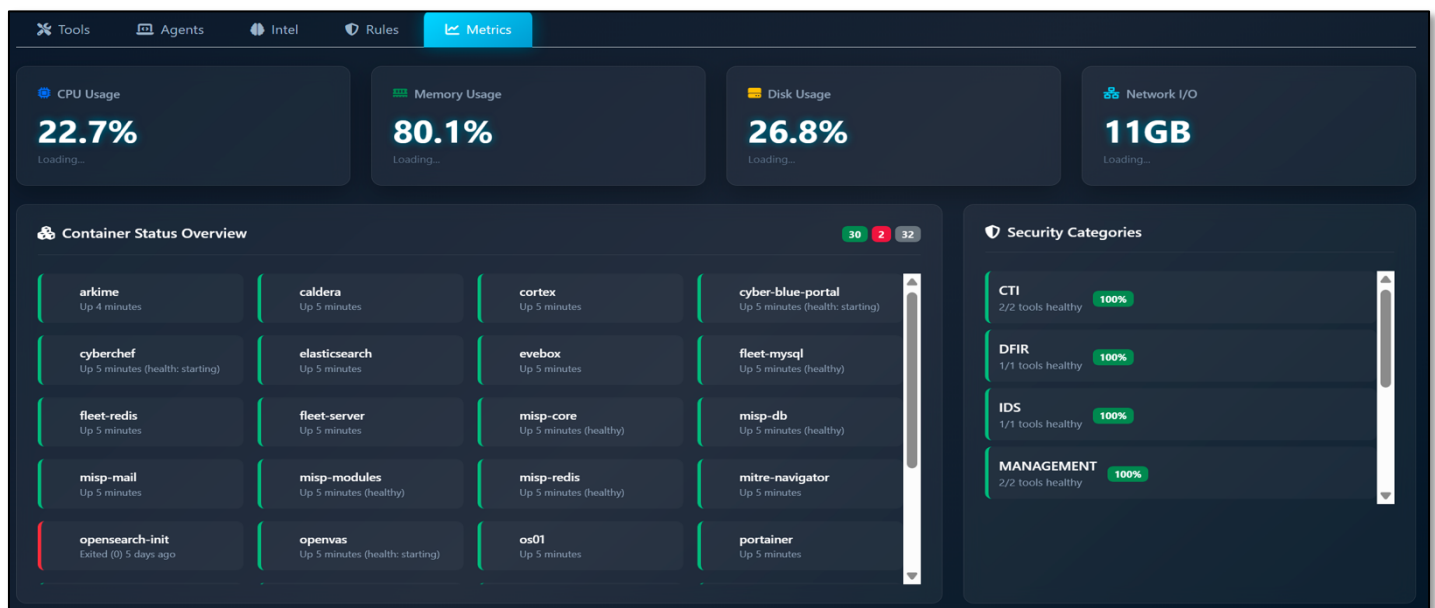


Figure 5: CyberBlueSOC Metrics Dashboard



Tool Cards Explained

Each security tool appears as a card with:

Header: Tool name + Category badge (SIEM/DFIR/CTI/SOAR)

Status: Green "Up X minutes" or Red "Stopped"

Description: What the tool does

Controls:

- [Open](#) - Launch tool in new tab
- [Credentials](#) - Show login info popup
- [Info](#) - Detailed information
- [Start](#) / [Restart](#) / [Stop](#) - Management buttons

The main dashboard displays multiple tool cards in a clean grid layout, each showing real-time status and instant access controls.

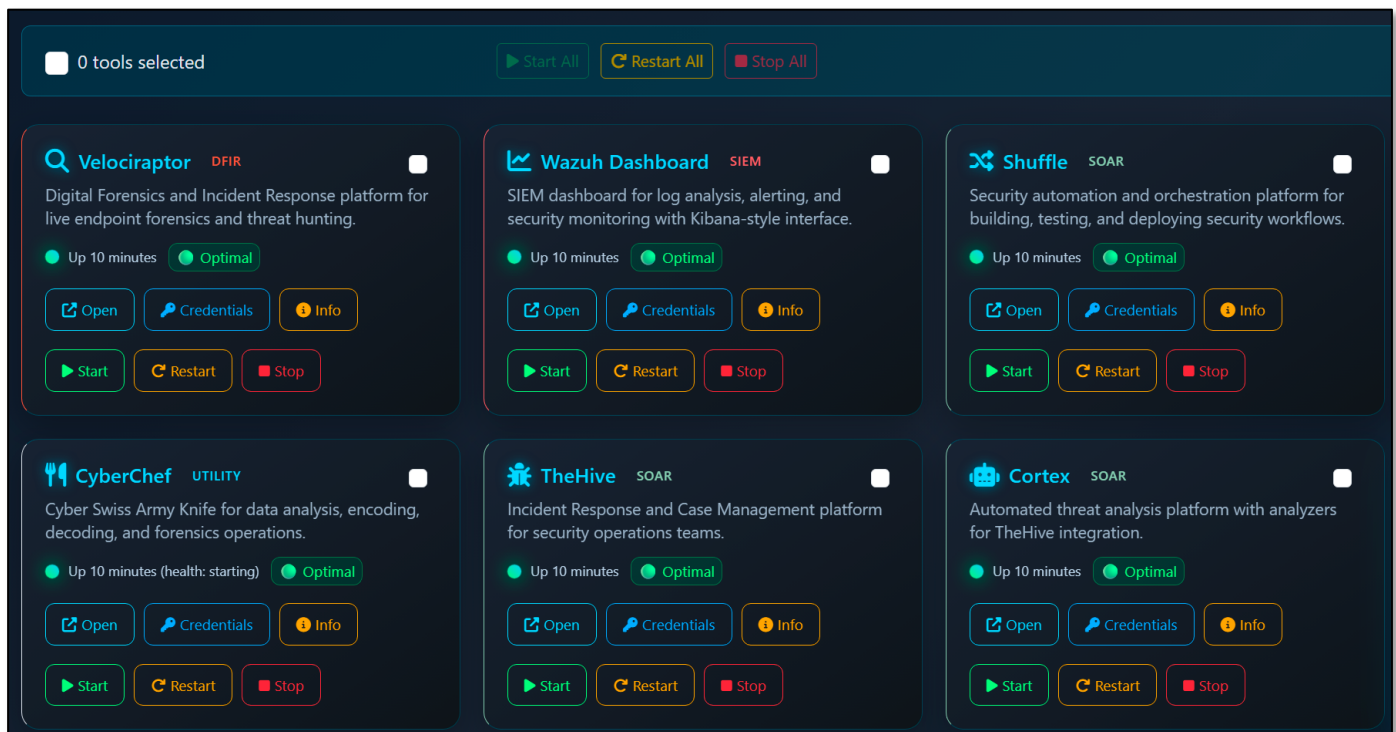


Figure 6: Tools Cards

Viewing Tool Credentials

Click the **Credentials** button on any tool card to see a popup with:

- Access URL (with copy button)
- Username and password
- Container status

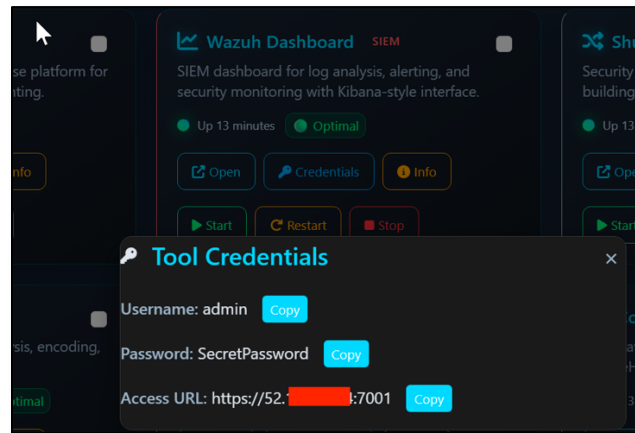


Figure 7: Tools Credentials

Viewing Tool Information

Click the **Information** button to see more details about the container status, Access, and quick start guide.

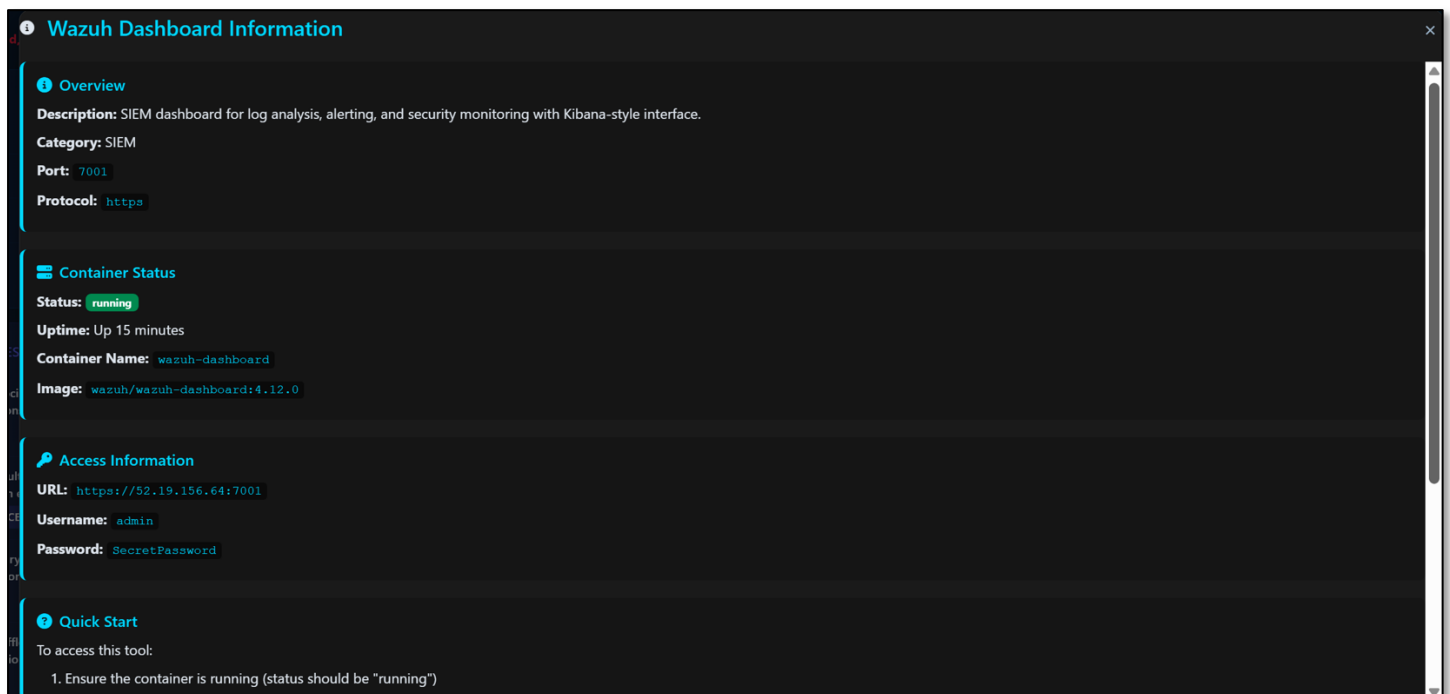


Figure 8: Tools Information feature



Tool Access Guide

All Your Tools (No Portal Password!)

QuickAccess

Portal has **no authentication**- just open and use!
https://YOUR_IP:5443 → Direct access to dashboard

Complete Tool List

Tool	URL	Credentials
CyberBlueSOC Portal	https://IP:5443	No password!
Velociraptor	https://IP:7000	admin / cyberblue
Wazuh	https://IP:7001	admin / SecretPassword
Shuffle	https://IP:7002	admin / password
MISP	https://IP:7003	admin@admin.test / admin
CyberChef	http://IP:7004	No auth
TheHive	http://IP:7005	admin@thehive.local / secret
Cortex	http://IP:7006	admin / cyberblue123
Fleet	http://IP:7007	Setup on first use
Arkime	http://IP:7008	admin / admin
Caldera	http://IP:7009	red / blue: cyberblue
Wireshark	https://IP:7099	admin / cyberblue
ATT&CK Navigator	http://IP:7013	No auth
EveBox	http://IP:7015	No auth
Portainer	https://IP:9443	admin / cyberblue123

Replace IP with your server's IP address from installation



Verify the Access to Tools

Velociraptor (EDR/DFIR)

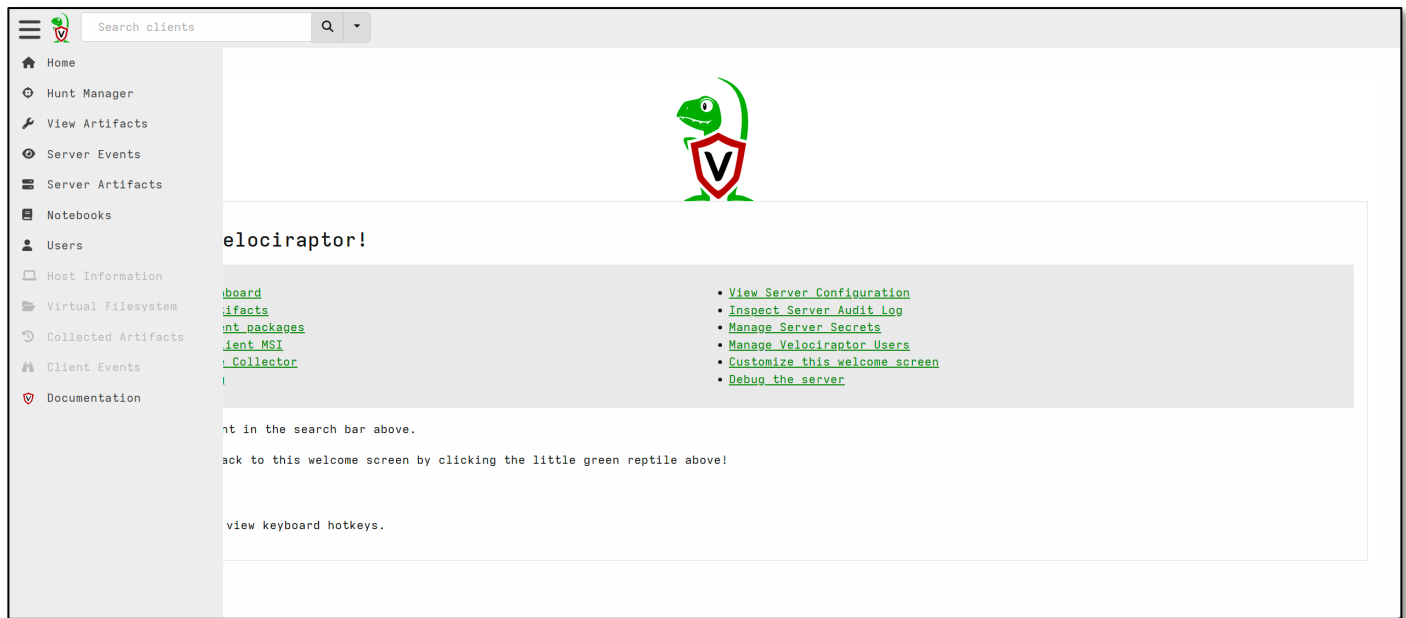


Figure 9: Velociraptor WebUI

Wazuh (SIEM)

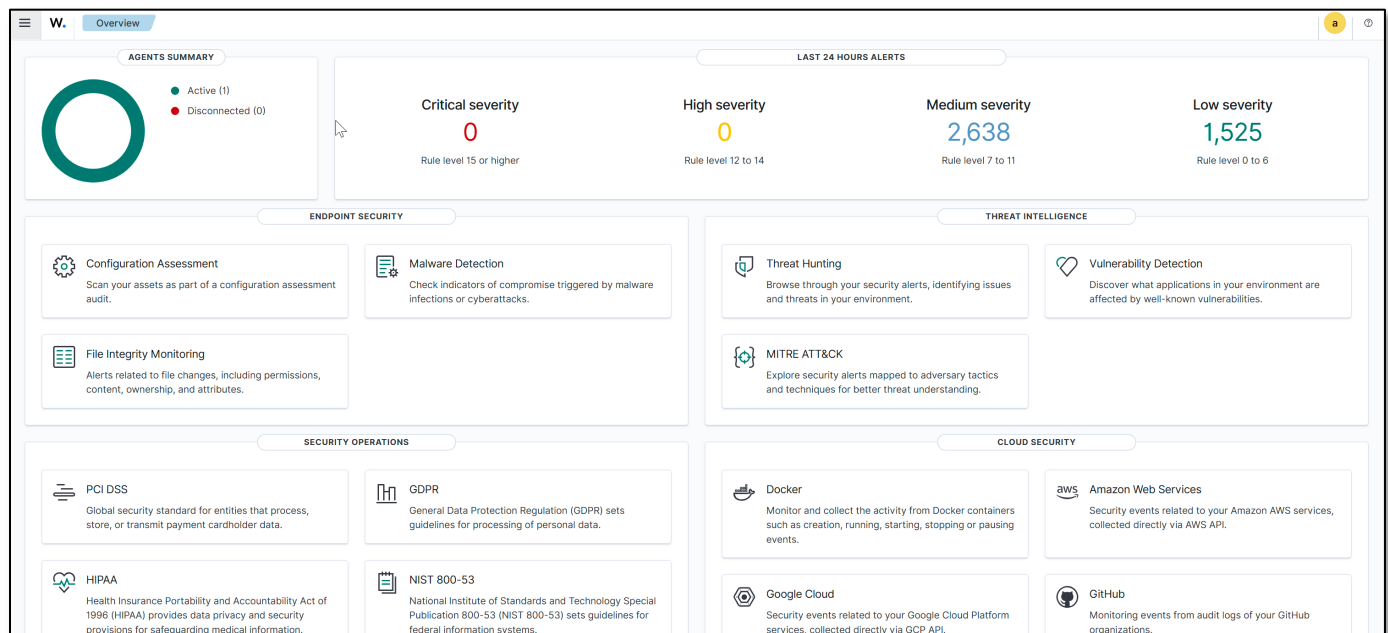


Figure 10: Wazuh WebUI

Shuffle (SOAR)

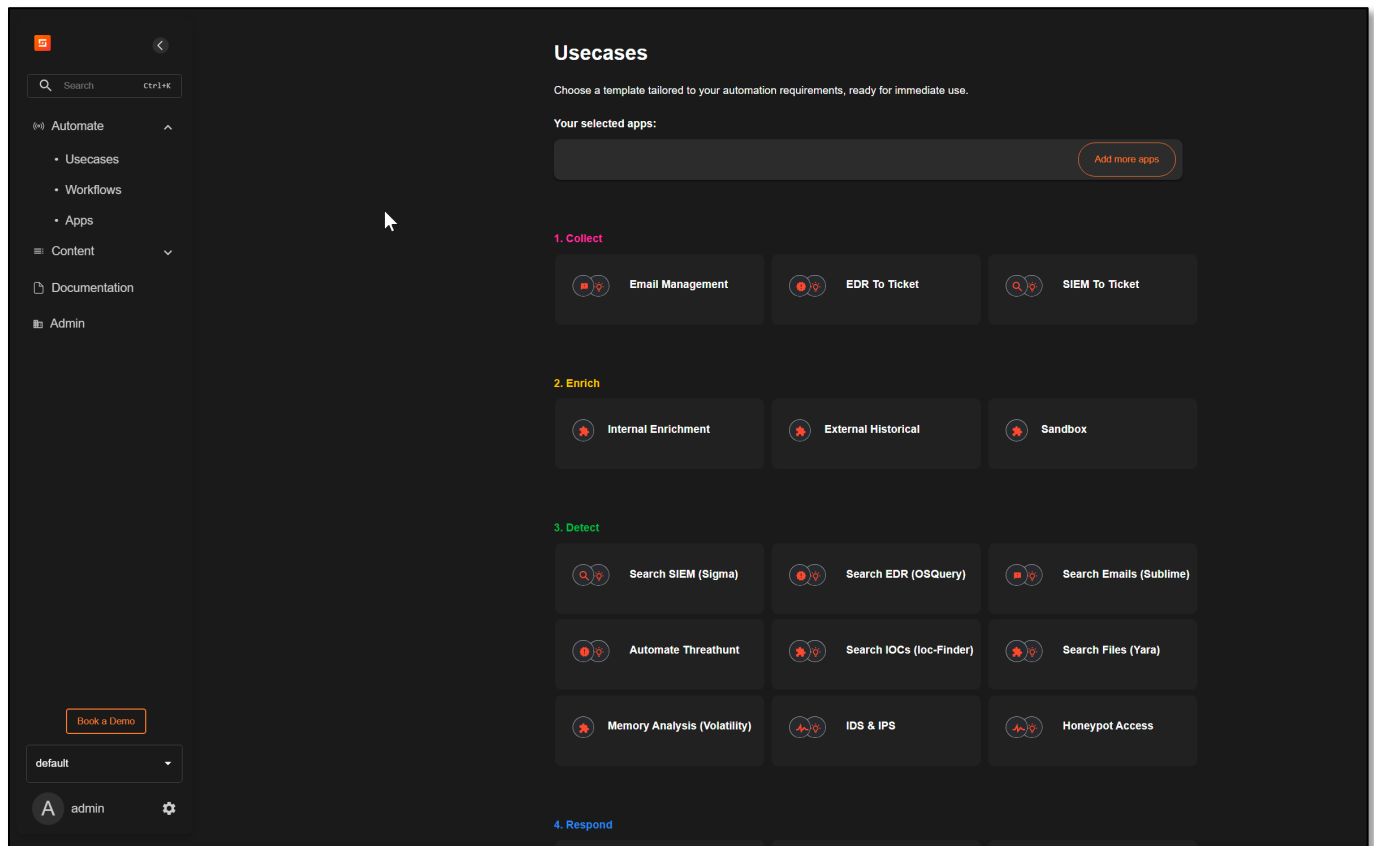


Figure 11: Shuffle WebUI

MISP (Threat Intel)

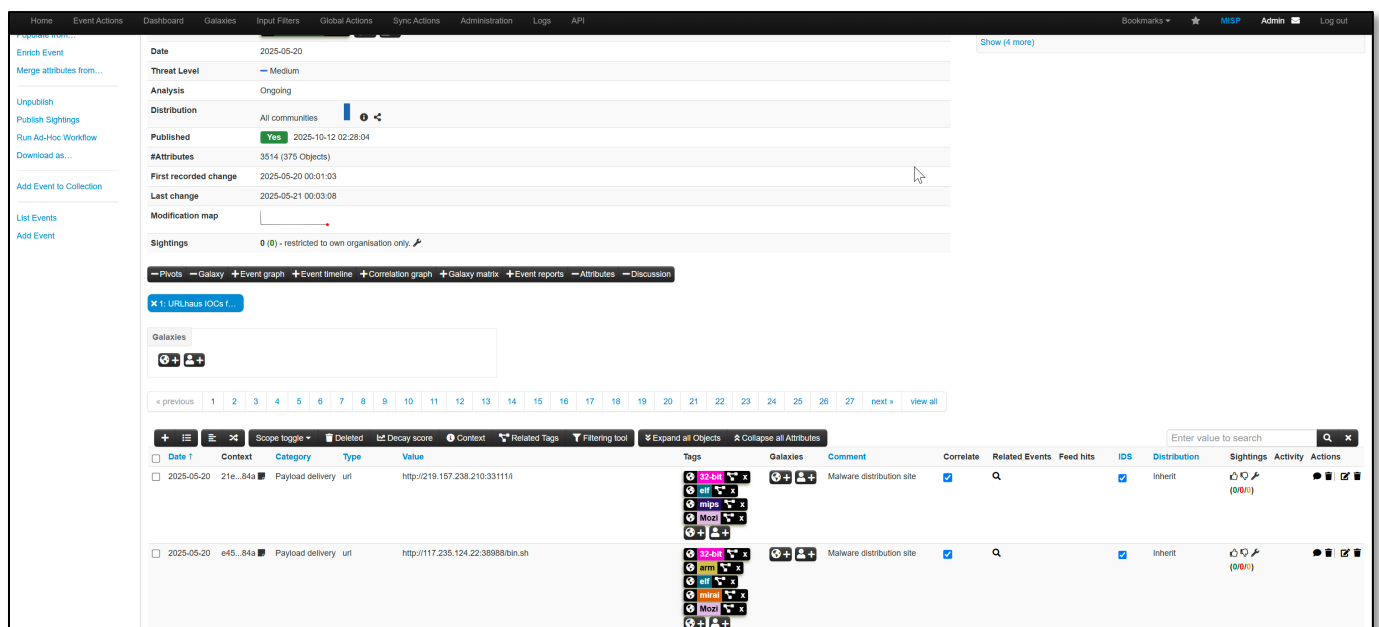


Figure 12: MISP WebUI

CyberChef (Utility: Swiss Army Knife)

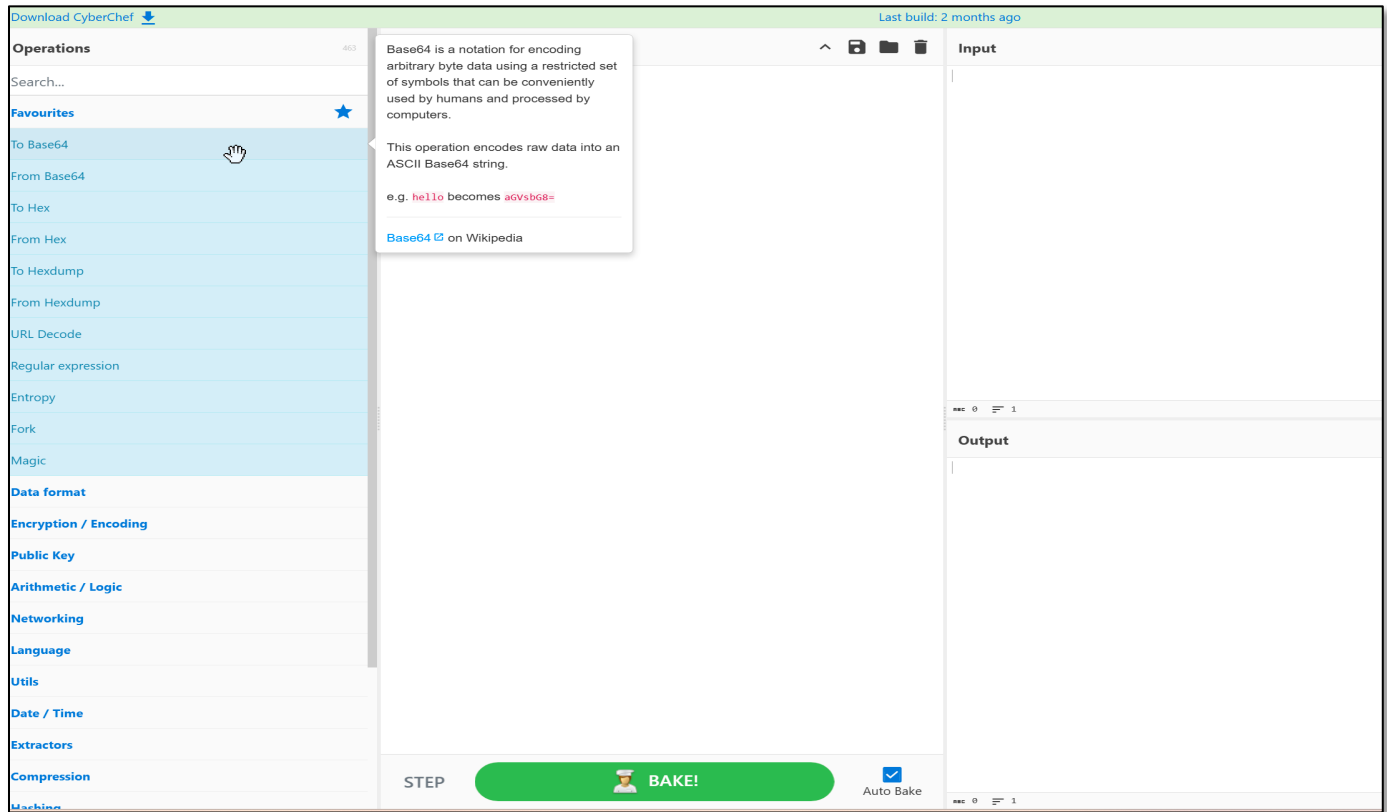


Figure 13: CyberChef WebUI

TheHive (Case Management)

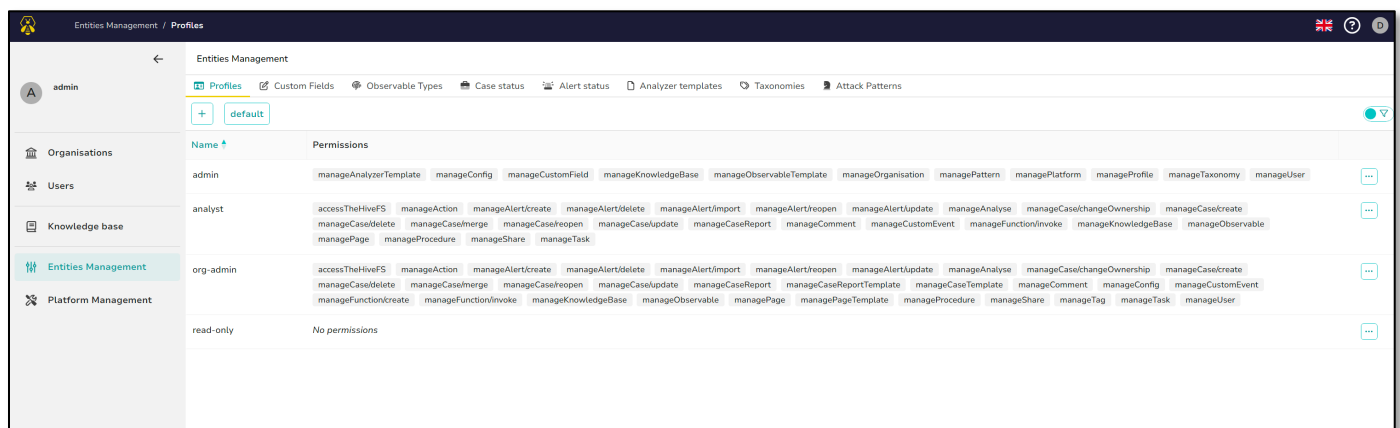


Figure 14: TheHive WebUI



Cortex (The Hive observables Analyzer)

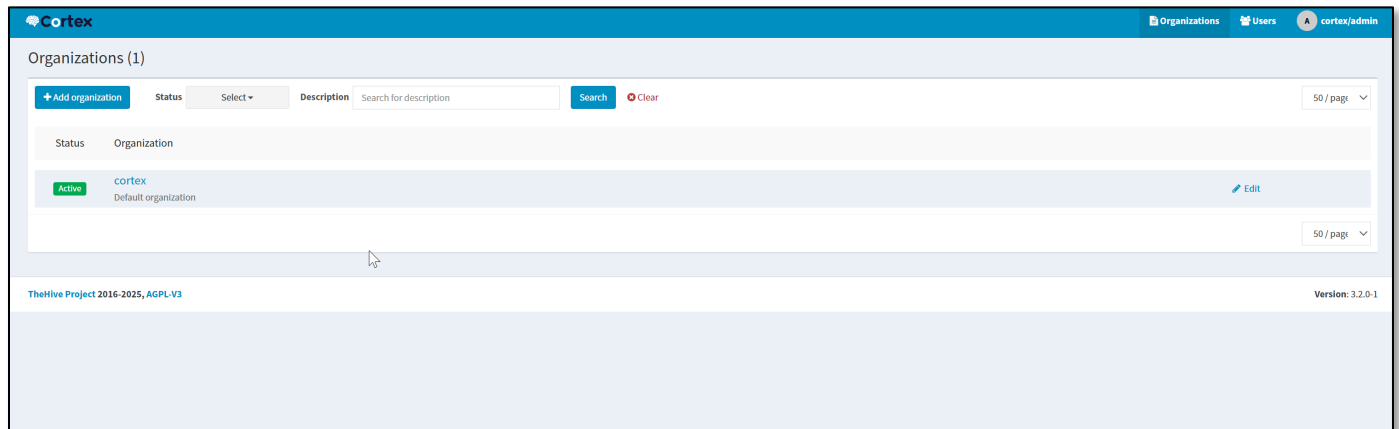


Figure 15: Cortex WebUI

Fleet (Device Management - Osquery)

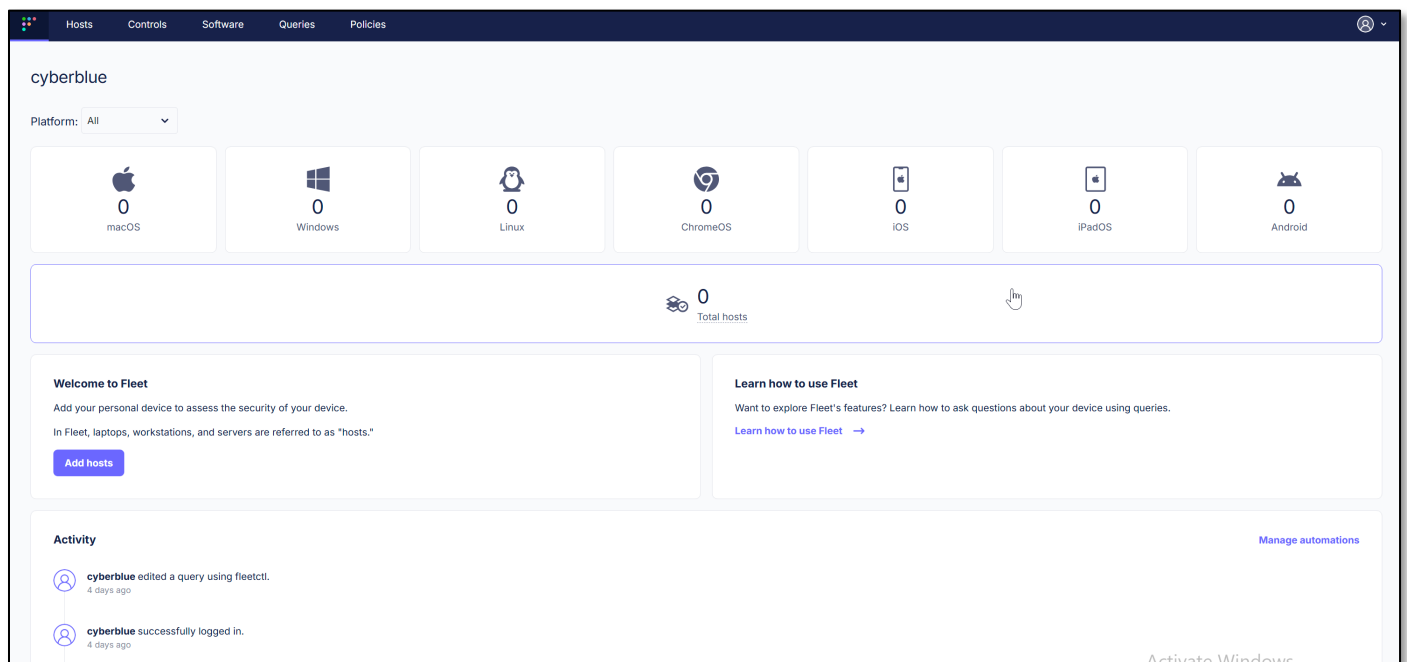


Figure 16: FleetDM WebUI



Arkime (packet capturing)

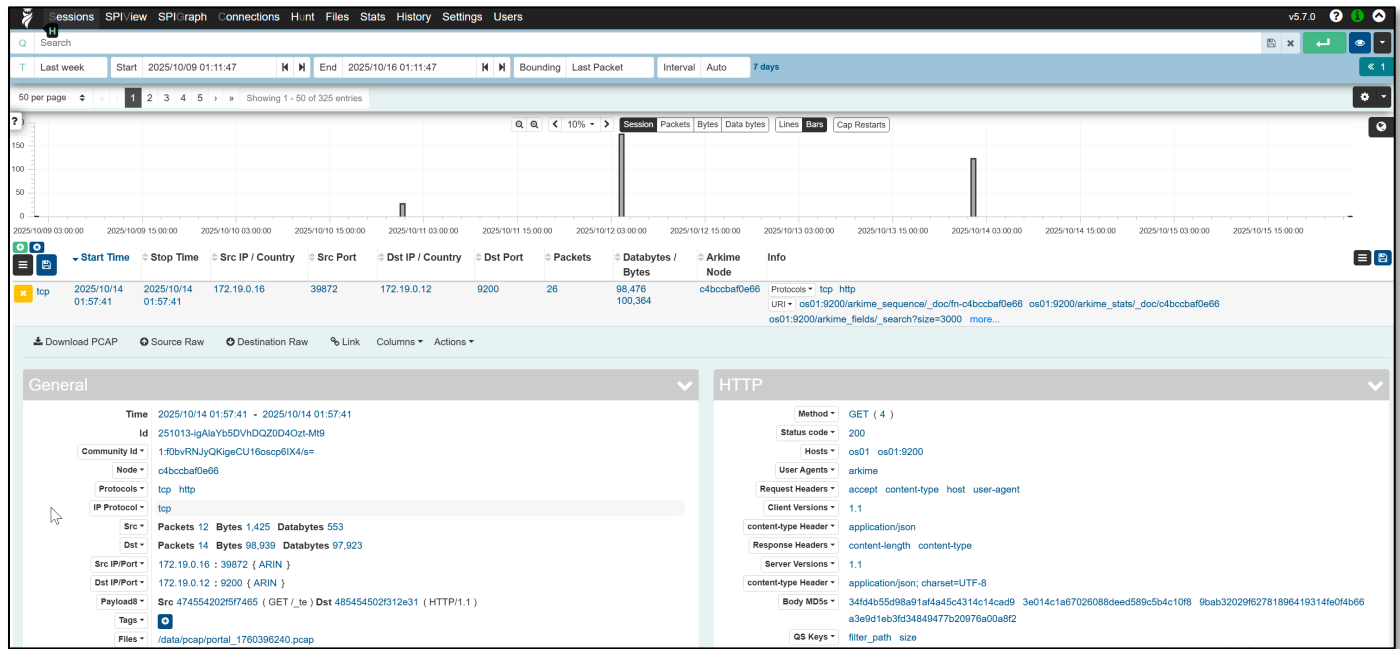


Figure 17: Arkime WebUI

MITRE Caldera (adversary emulation)

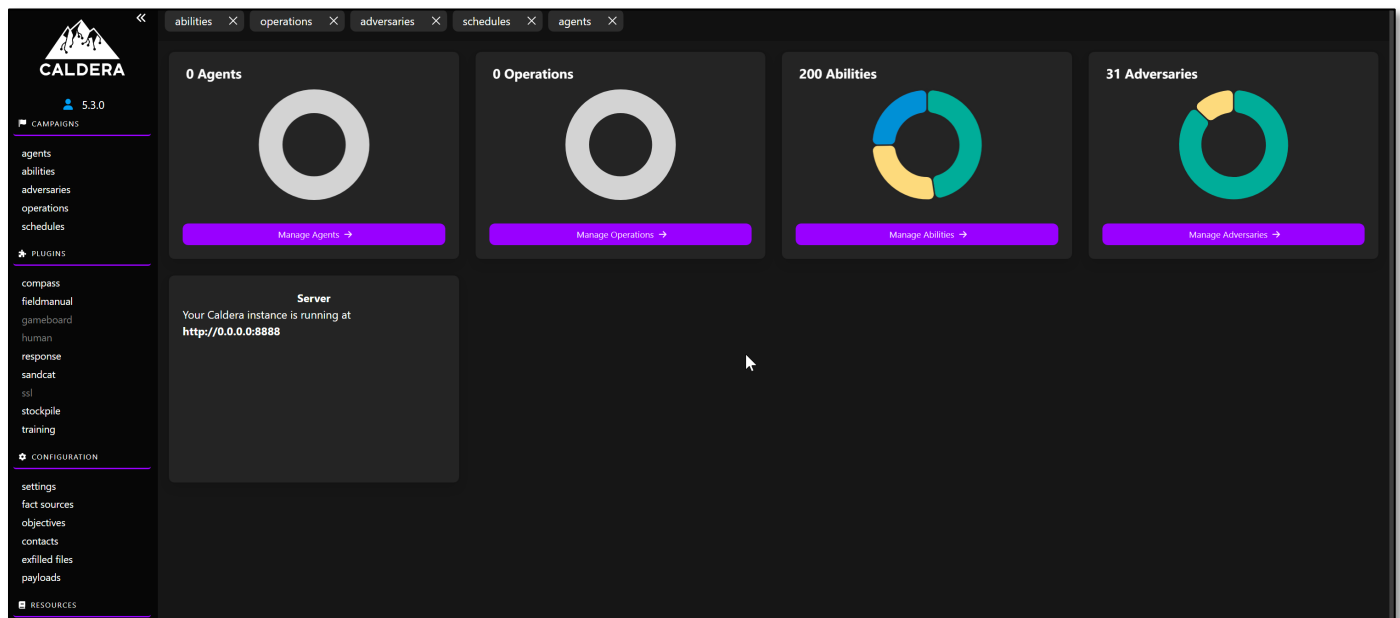


Figure 18: MITRE Caldera WebUI



EveBox (Suricata alert and event management)

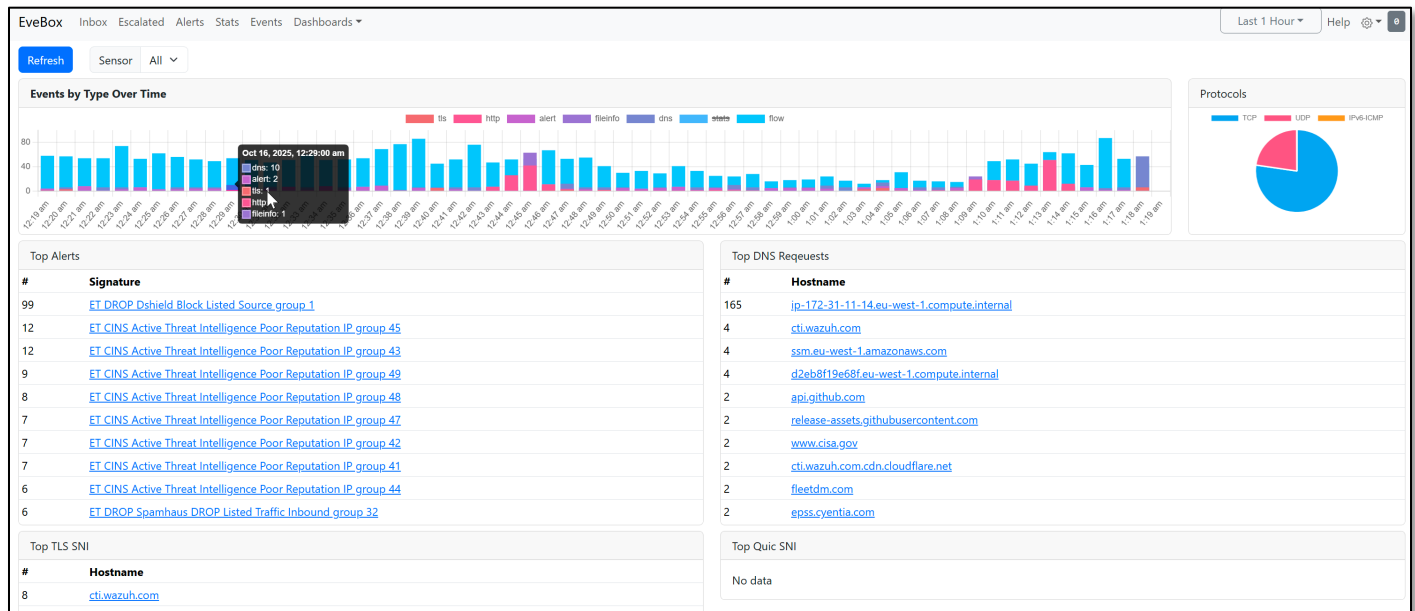


Figure 19: EveBox WebUI

Wireshark (Network Analysis)

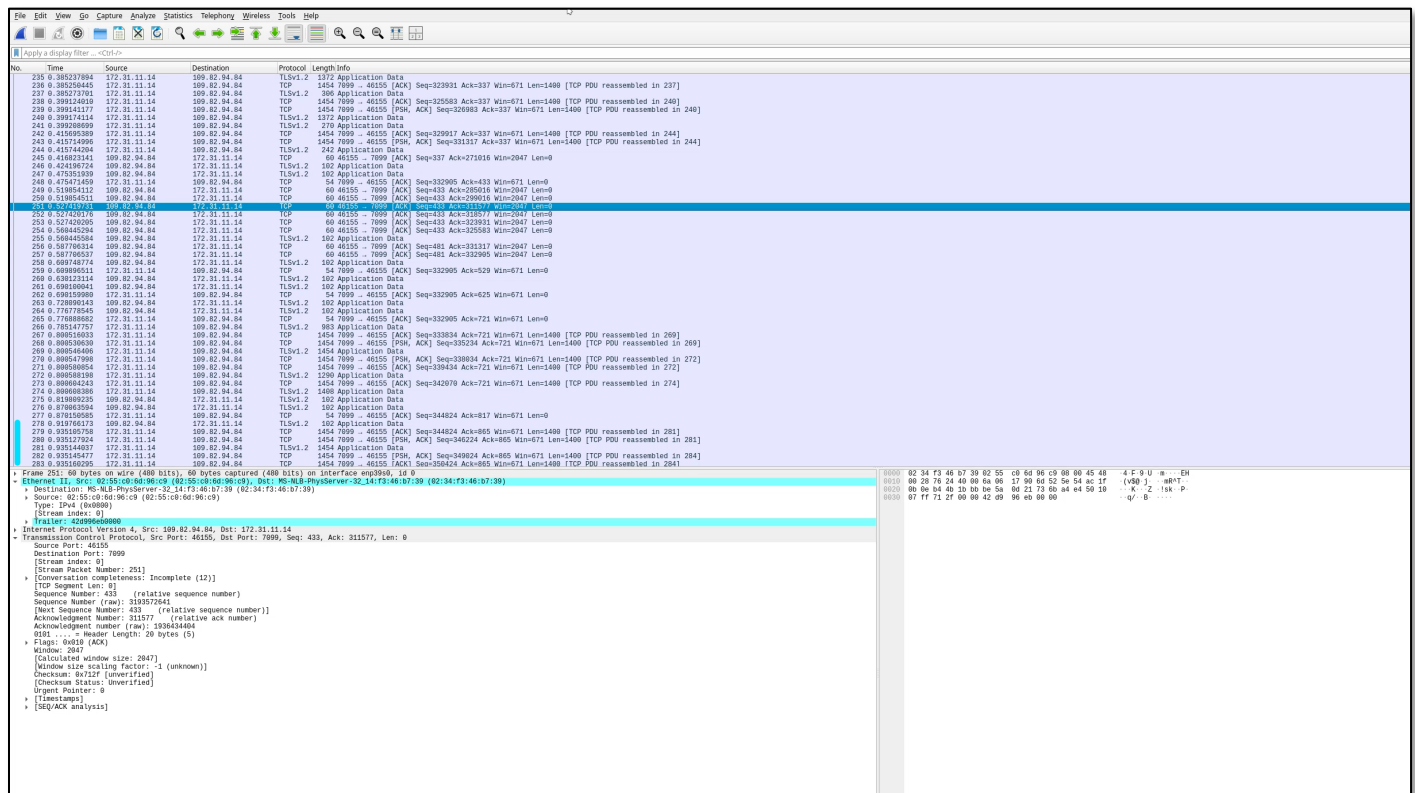


Figure 20: Wireshark WebUI



ATT&CK Navigator (ATT&CK matrices)

The screenshot displays the MITRE ATT&CK Navigator WebUI. The interface features a top navigation bar with 'Selection Controls', 'Layer Controls', and 'Technique Controls'. Below this, a grid of attack techniques is presented, organized into columns: Initial Access (8 techniques), Execution (4 techniques), Persistence (7 techniques), Privilege Escalation (3 techniques), Defense Evasion (17 techniques), Credential Access (5 techniques), Discovery (8 techniques), Lateral Movement (2 techniques), Collection (13 techniques), Command and Control (9 techniques), Exfiltration (2 techniques), and Impact (10 techniques). The 'Defense Evasion' column is currently selected, and the 'Impair Defenses' technique is highlighted. The 'Impair Defenses' technique is described as 'Impair Defenses (T1629)' and is associated with the 'Defense Evasion' category. The 'Impair Defenses' technique is further detailed as 'Impair Defenses (T1629)' and is associated with the 'Defense Evasion' category.

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
Application Versioning	Command and Scripting Interpreter	Boot or Logon Initialization Scripts	Abuse Elevation Control Mechanism	Application Versioning	Access Notifications	File and Directory Discovery	Exploitation of Remote Services	Access Notifications	Application Layer Protocol	Exfiltration Over Alternative Protocol	Account Access Removal
Drive-By Compromise	Exploitation for Client Execution	Compromise Application Executable	Exploitation for Privilege Escalation	Download New Code at Runtime	Clipboard Data	Location Tracking	Replication Through Removable Media	Adversary-in-the-Middle	Call Control	Exfiltration Over C2 Channel	Call Control
Exploitation for Initial Access	Native API	Compromise Client Software Binary	Process Injection	Execution Guardrails	Credentials from Password Store	Network Service Scanning		Archive Collected Data	Dynamic Resolution		Data Destruction
Lockscreen Bypass	Scheduled Task/Job	Event Triggered Execution		Foreground Persistence	Input Capture	Process Discovery		Audio Capture	Encrypted Channel		Data Encrypted for Impact
Phishing		Foreground Persistence		Hide Artifacts	Steal Application Access Token	Software Discovery		Call Control	Ingress Tool Transfer		Data Manipulation
Replication Through Removable Media		Hijack Execution Flow		Hooking	Impair Defenses (T1629)	System Information Discovery		Clipboard Data	Non-Standard Port		Endpoint Denial of Service
SIM Card Swap		Scheduled Task/Job		Indicator Removal on Host		System Network Configuration Discovery		Data from Local System	Out of Band Data		Generate Traffic from Victim
Supply Chain Compromise				Input Injection		System Network Connections Discovery		Input Capture	Remote Access Software		Input Injection
				Masquerading				Location Tracking	Web Service		Network Denial of Service
				Native API				Protected User Data			SMS Control
				Obfuscated Files or Information				Screen Capture			
				Process Injection				Stored Application Data			
				Proxy Through Victim				Video Capture			
				Subvert Trust Controls							
				Virtualization Solution							
				Virtualization/Sandbox Evasion							

Figure 21: MITRE ATT&CK Navigator WebUI

Portainer (containerized applications)

The screenshot displays the Portainer WebUI. The interface features a left sidebar with navigation options: Home, local, Dashboard, Templates, Stacks, Containers, Images, Networks, Volumes, Events, Host, Administration, User-related, Environment-related, Registries, Logs, Notifications, and Settings. The main content area shows the 'Environment summary' and 'Dashboard'. The 'Environment summary' section includes details about the environment: local @ 8 @ 16.5 GB - Standalone 28.5.1. The 'Dashboard' section provides a summary of the environment's state: 1 Stack, 46 Images, 5 Networks, 32 Containers (30 running, 2 stopped, 0 healthy, 0 unhealthy), and 24 Volumes.

Environment summary	Dashboard
Environment	local @ 8 @ 16.5 GB - Standalone 28.5.1
URL	/var/run/docker.sock
GPU	none
Tags	-
Stack	1
Images	46
Networks	5
Containers	32
Volumes	24

Figure 22: Portainer WebUI

Deploying Agents

Agent Deployment Center

See the following Screenshot for Agents tab showing all deployment options

The Agents tab provides one-click deployment packages for:

Velociraptor (DFIR Agent)

1. Download pre-configured packages for:
 - Windows (velociraptor-windows.exe)
 - Linux (velociraptor-linux)
 - macOS Intel (velociraptor-macos-intel)
 - macOS ARM (velociraptor-macos-arm)
2. Install on target endpoint using the installation guide in the package.
3. Agent auto-connects to your server!

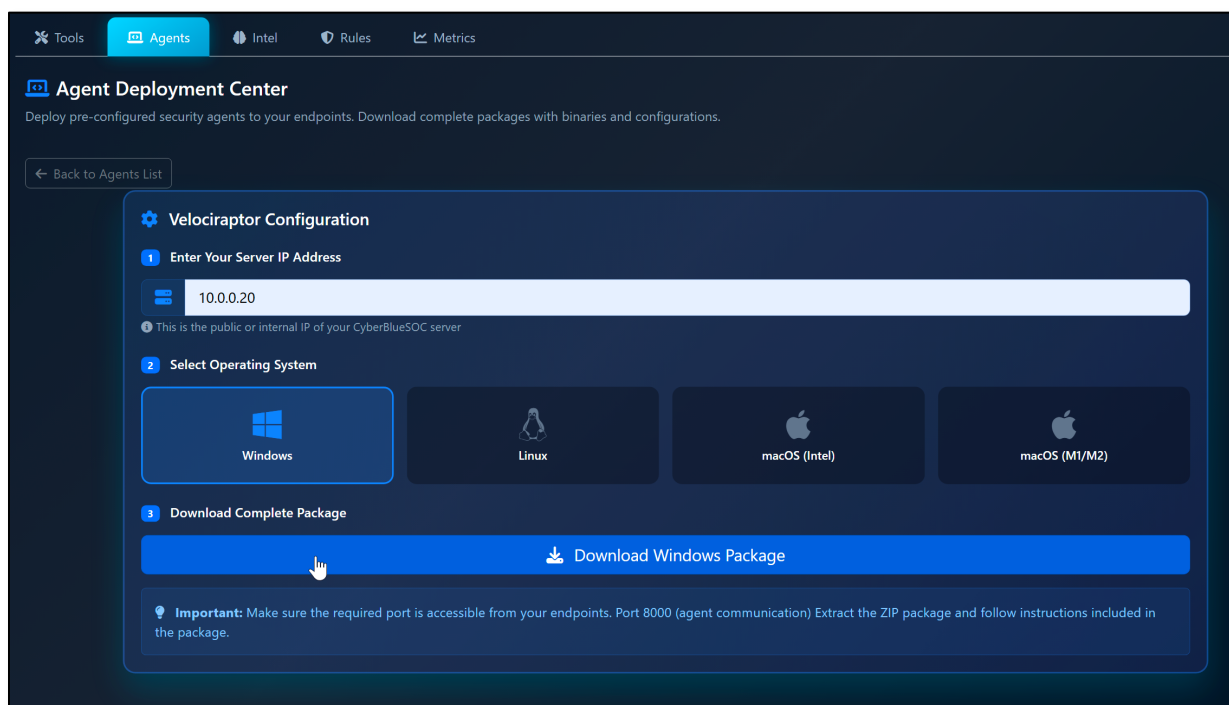


Figure 23: Velociraptor Agent Deployment

Wazuh (SIEM Agent)

The following screenshot shows the configuration wizard Steps:

1. Enter your server IP (e.g., 10.0.0.20)
2. Select OS: Windows / Ubuntu-Debian / RHEL-CentOS
3. Click "Download Windows Package" (or Linux/macOS)
4. Install on target endpoint using the installation guide in the package.
5. Agent auto-connects to your server!

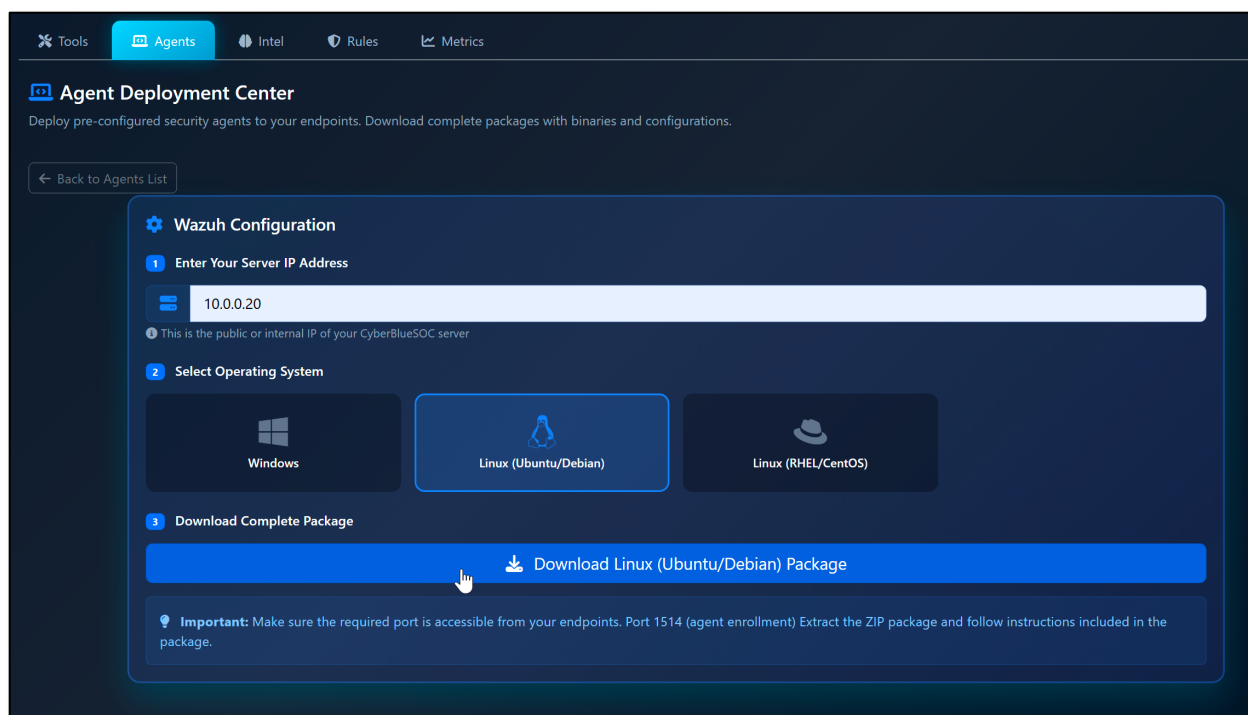


Figure 24: Wazuh Agent Deployment

Arkime Network Capture (PCAP)

1. Select Arkime PCAP from the Agents tab
2. Select the Capture duration and start capturing traffic
3. Open in Arkime to see the captured packets

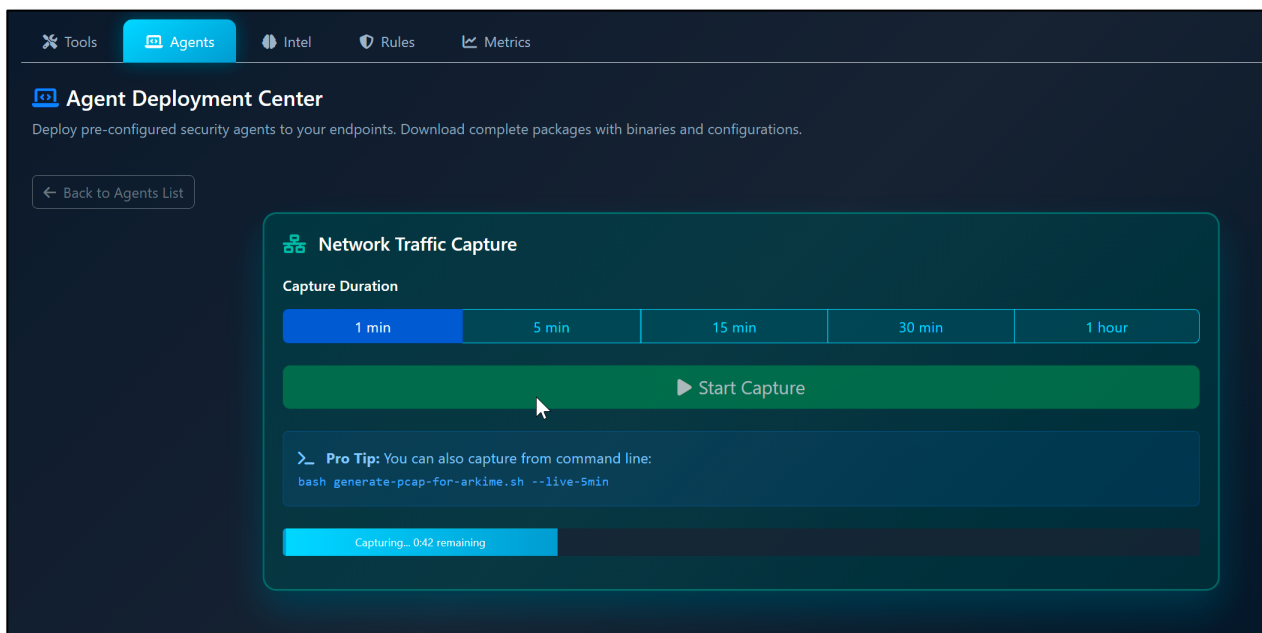


Figure 25: Arkime PCAP Generator



Threat Intelligence

IOC Search Interface

See the following screenshot Intel tab with search and results

Search 250,000+ threat indicators instantly:

Search for:

- IP addresses
- Domain names
- File hashes (MD5, SHA1, SHA256)
- URLs

Test buttons included:

- Test: IP (pre-loaded malicious IP)
- Test: Domain (example domain)
- Test: Hash (malware hash)

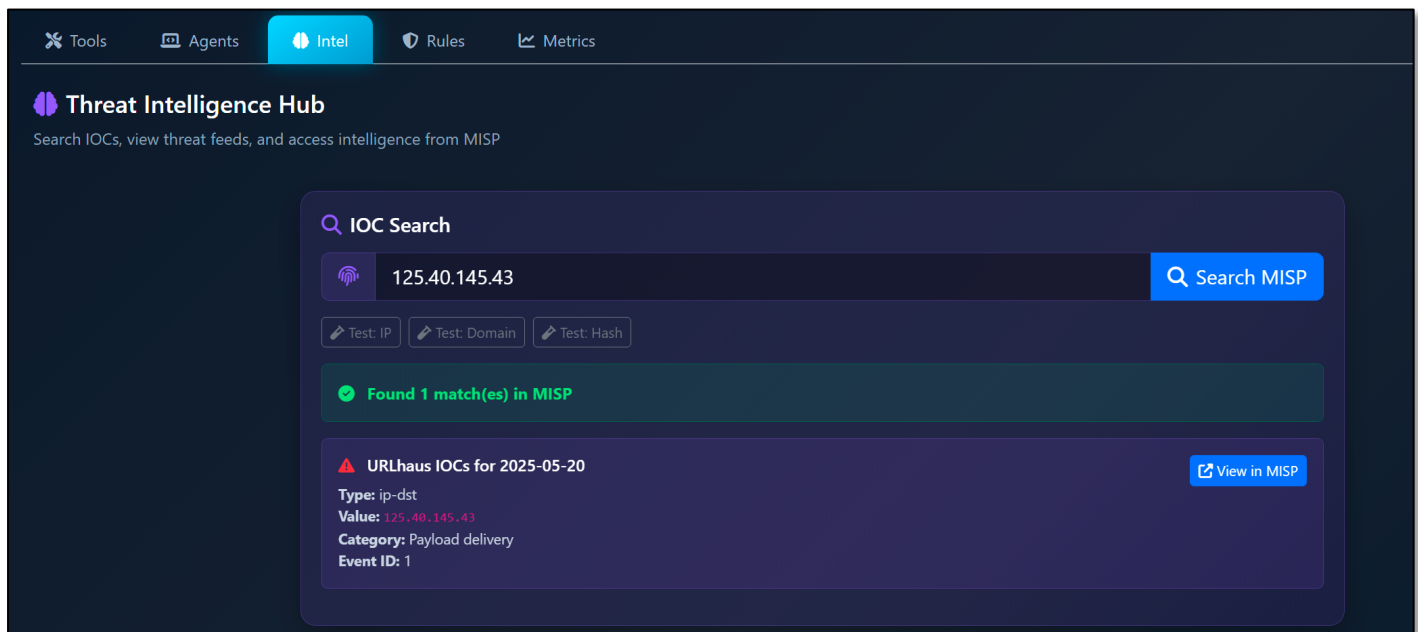


Figure 26: MISP IOC Search



MISP Statistics Dashboard

Visible in the following screenshot

Live metrics:

- h. **20 Events** - Threat Event groups/campaigns
- i. **250k+ Attributes** - Individual IOCs
- j. **4 Active Feeds** - Auto-updating sources

Click "Open MISP Dashboard" for full threat intelligence interface.

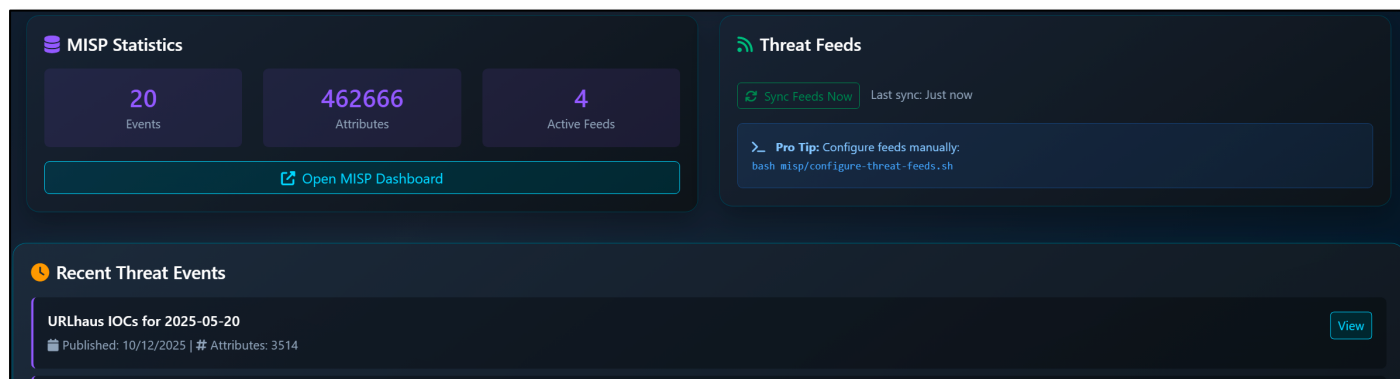


Figure 27: MISP statistics



Detection Rules

YARA Malware Detection

See the following Screenshots, Complete rules browser interface

523 rules across categories:

Category	Rules
malware	363
deprecated	66
maldocs	20
exploit_kits	11
packers	6
webshells	9
cve_rules	14
crypto	1

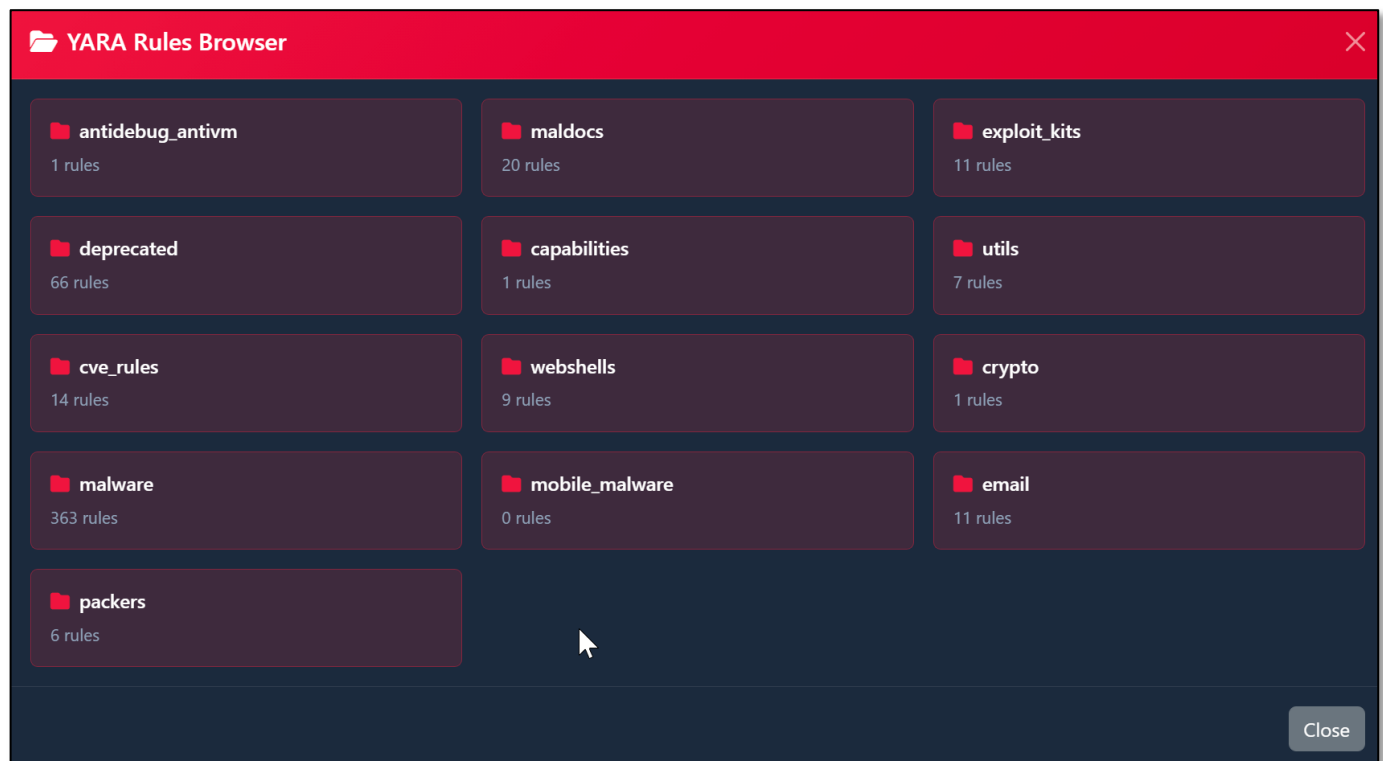


Figure 28: YARA Rules Categories browser



The following screenshot shows the YARA browser with a red-themed interface displaying all categories as clickable tiles.

Browsing rules:

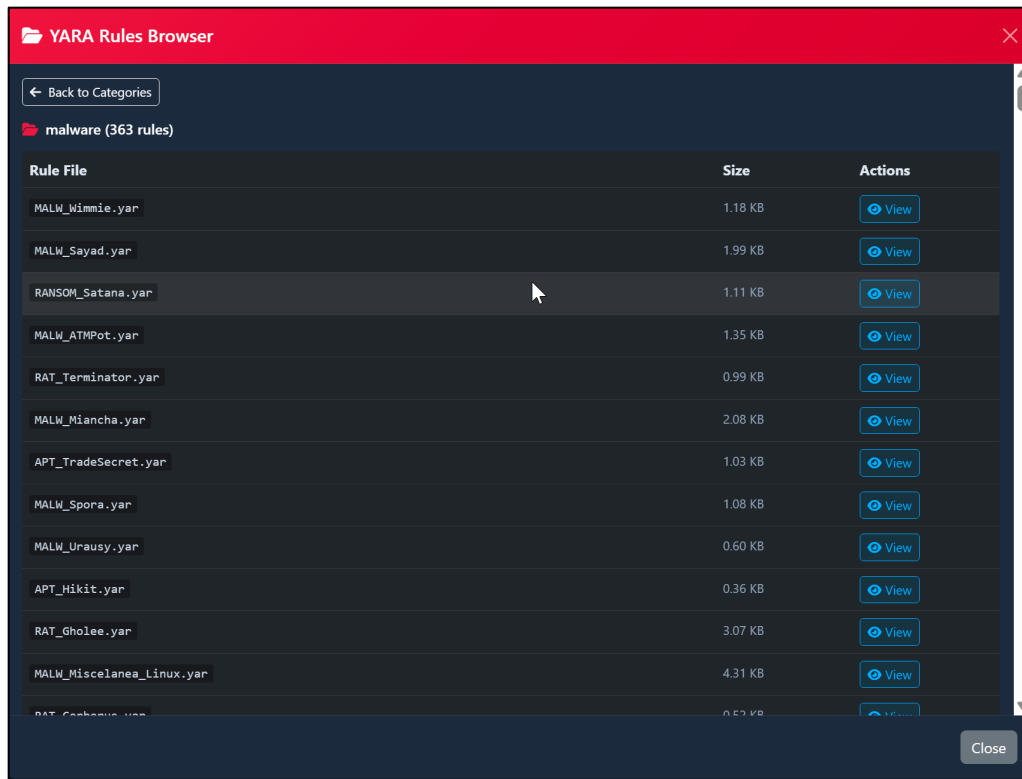


Figure 29: YARA Rules Browser

Viewing rule content:

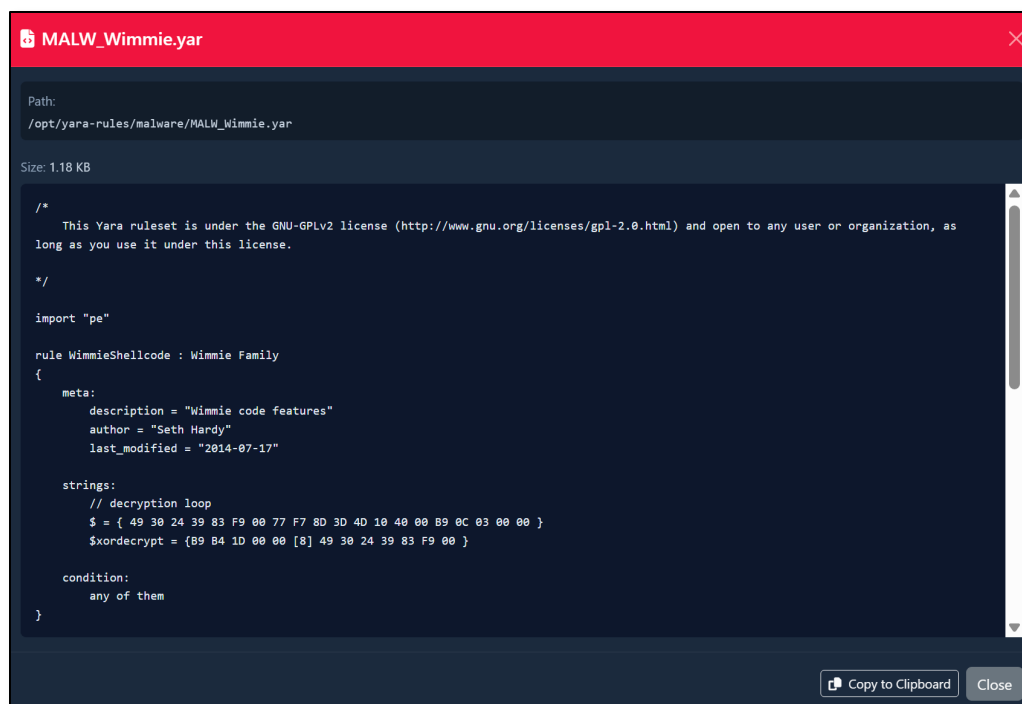


Figure 30: YARA Rule Content



Sigma Detection Rules

3,047 rules for multiple platforms:

Platform	Rules
windows	2,338
cloud	269
linux	206
macos	89
application	63
network	47
web	45

The following screenshots shows the Sigma browser with a blue-themed interface. The next Screenshot displays cloud- specific rules like `aws_enum_buckets.yml` for detecting bucket enumeration attacks on AWS CloudTrail.

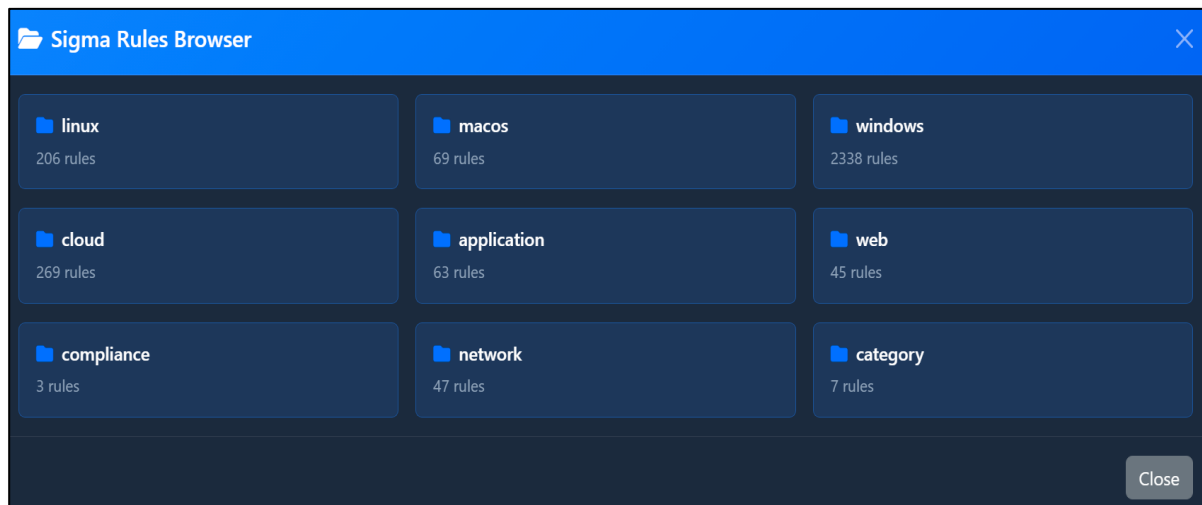


Figure 31: Sigma Rules Categories

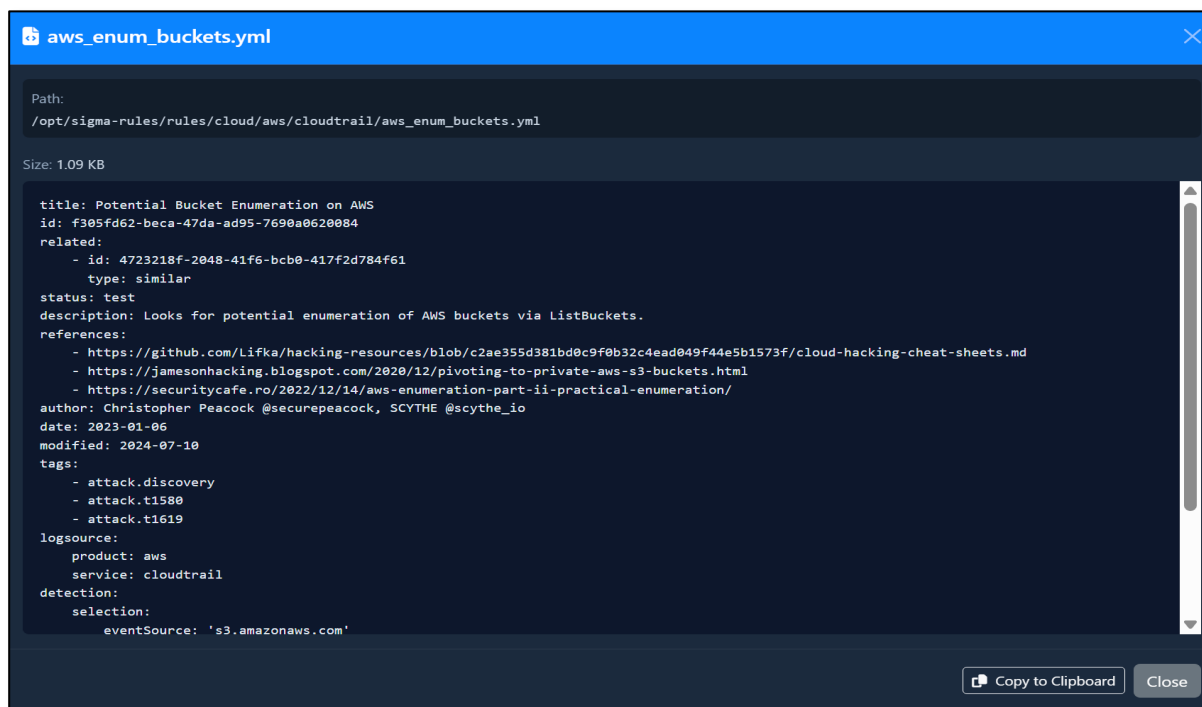


Figure 32: Sigma Rule Content



System Monitoring

Metrics Dashboard

See the following Screenshot: Complete system metrics interface

Four key performance gauges:

- **CPU Usage:** Real-time processor utilization
- **Memory Usage:** RAM consumption monitoring
- **Disk Usage:** Storage space tracking
- **Network I/O:** Total data transfer

Container Status Grid:

Visual overview of all 30+ containers:

- Green = Running healthy
- Red = Stopped
- Each container shows name and uptime

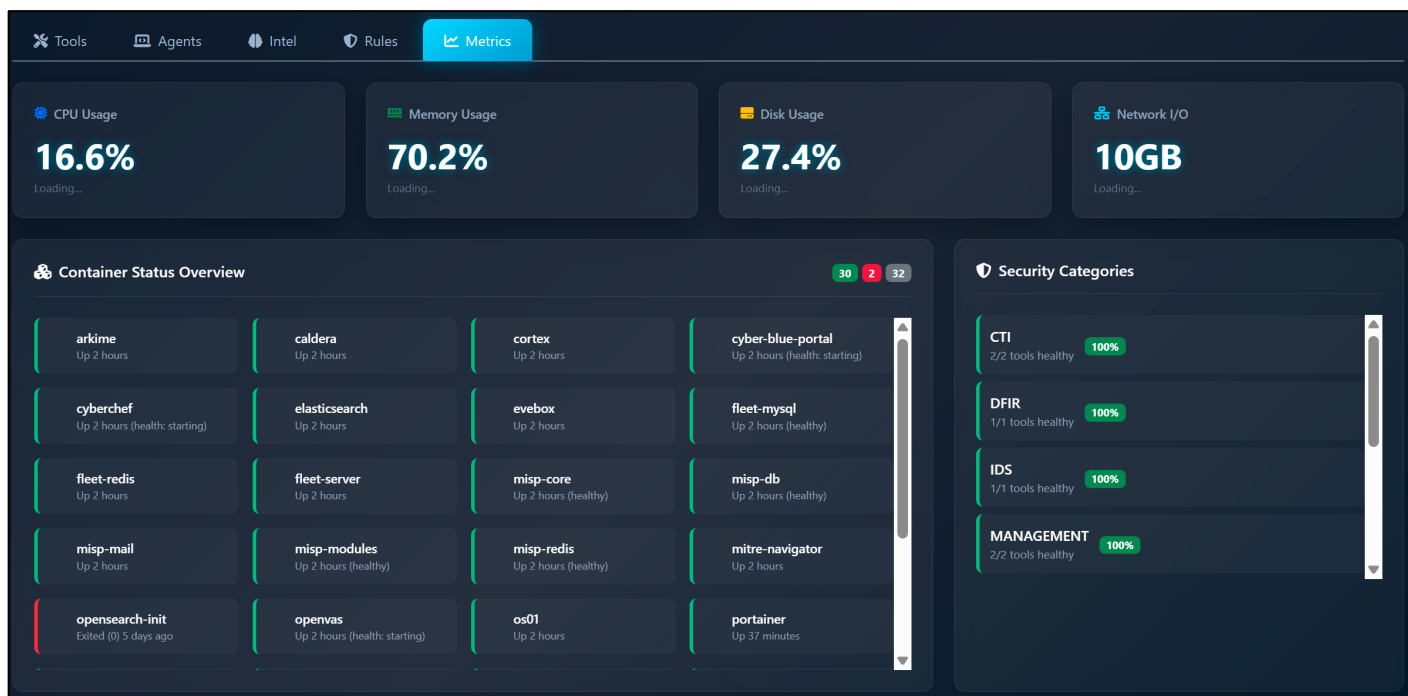


Figure 33: CyberBlueSOC Metrics Dashboard



Troubleshooting

All tools stopped?

```
./force-start.sh
```

Check logs:

```
sudo docker logs [container-name]
```

Restart specific tool:

```
sudo docker-compose restart [service-name]
```

Get back to the following full troubleshooting guide:

<https://github.com/cyberblu3s/CyberBlue/blob/main/docs/TROUBLESHOOTING.md>

Quick Reference

Essential Commands

Check all containers

```
sudo docker ps
```

Restart all services

```
sudo docker-compose restart
```

Emergency restart

```
./force-start.sh
```

Update threat feeds

```
bash misp/configure-threat-feeds.sh
```

System Locations

YARA Rules: /opt/yara-rules/

Sigma Rules: /opt/sigma-rules/

Project: ~/CyberBlue/



Common Issues & Solutions

Reference: Get back to the following full troubleshooting guide:

<https://github.com/cyberblu3s/CyberBlue/blob/main/docs/TROUBLESHOOTING.md>

Installation Problems

Port Already in Use

Problem: Error message: "bind: address already in use"

Solution:

Find what's using the port

```
sudo lsof -i :7001
```

Kill the process

```
sudo kill -9 [PID]
```

OR change port in docker-compose.yml

Out of Disk Space

Problem: "no space left on device"

Solution:

Check space

```
df -h
```

Clean Docker

```
sudo docker system prune -a -f
```

Remove old images

```
sudo docker image prune -a -f
```

Docker Permission Denied

Problem: "permission denied while connecting to Docker daemon"

Solution:

Add user to docker group

```
sudo usermod -aG docker $USER
```

Re-login or run:

```
newgrp docker
```

Verify

```
docker ps
```



Runtime Issues

Container Keeps Restarting

Problem: Container shows "Restarting" status

Solution:

Check logs

```
sudo docker logs [container-name]
```

Common fixes:

1. Memory issue - check swap

```
free -h swapon
```

```
--show
```

2. Config error - restart container

```
sudo docker-compose restart [container-name]
```

3. Dependency issue - restart all

```
./force-start.sh
```

Portal Not Loading

Problem: Can't access https://IP:5443

Solution:

Check portal container

```
sudo docker ps | grep portal
```

Restart portal

```
sudo docker-compose restart portal
```

Check logs

```
sudo docker logs cyber-blue-portal
```

Verify port not blocked

```
sudo ss -tulpn | grep 5443
```



Tool Shows "Stopped"

Problem: Tool card shows red "Stopped" status

Quick Fix: - Click green "Start" button on tool card - Wait 30 seconds - Refresh page

If that doesn't work:

```
sudo docker-compose restart [service-name]
```

Performance Issues

High Memory Usage

Problem: System using >90% memory

Solutions:

```
# Check memory
```

```
free -h
```

```
# Restart memory-heavy services
```

```
sudo docker-compose restart wazuh-indexer
```

```
sudo docker-compose restart shuffle-opensearch
```

```
# Add more swap if needed
```

```
sudo fallocate -l 16G /swapfile2
```

```
sudo chmod 600 /swapfile2
```

```
sudo mkswap /swapfile2
```

```
sudo swapon /swapfile2
```

Slow Portal Response

Problem: Portal feels sluggish

Solutions: 1. Check container count:

```
sudo docker ps | wc -l      # Should be 30+
```

2. Restart Docker:

```
./force-start.sh
```

3. Check system resources in Metrics tab



MISP Issues

No Threat Indicators Problem: MISP shows 0 attributes

Solution:

```
# Manual feed sync
cd ~/CyberBlue
bash misp/configure-threat-feeds.sh

# Check logs
tail -f /var/log/misp-feed-sync.log

# Wait 10-15 minutes for download
```

Can't Login to MISP

Problem: MISP requires password change

Solution:

```
# Already configured - use: # admin@admin.test / admin
# If still issues, reset:
cd ~/CyberBlue
bash force-create-misp-admin.sh
```

Network Issues

Suricata Not Capturing Problem: EveBox shows no events

Solution:

```
# Update network interface
cd ~/CyberBlue
./update-network-interface.sh --restart-suricata
# Verify interface
ip route | grep default
# Check Suricata logs
sudo docker logs suricata
```



Can't Access Tools from Remote

Problem: Tools work on localhost but not from other computers

Solution:

```
# Check firewall (for cloud deployments) # AWS: Update Security Group
# Azure: Update Network Security Group # GCP: Update Firewall Rules
# For Ubuntu UFW:
sudo ufw allow 5443/tcp
sudo ufw allow 7000:7099/tcp
sudo ufw allow 9443/tcp
sudo ufw enable
```

CyberBlueSOC!

Your SOC & Blue team journey can starts now!
Happy hunting!

CyberBlueSOC v1.0 • Educational Use Only • MIT License

<https://github.com/cyberblu3s/CyberBlue>